Harsh Patel

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SKILLS

Firmware Arduino, STM Nucleo, Raspberry Pi, Breadboarding, Soldering, Switch Logic, Displays

Languages C++, Python, Javascript/Typescript, HTML, CSS, Java, Processing

Tools Arduino IDE, Git, Github, VS-Code, UNIX Command-Line

Other Technical Documentation, Agile/Scrum Methodologies, Solution Architecture, Machine Learning

Soft Collaboration, Communication, Leadership, Time Management, Organization, Fast Learner

EXPERIENCE

Software Engineer - Friendlier Jan 2022 - Apr 2022

- Developed an automated payment and invoicing system using the Stripe API saving the company 32 hours per month
- Constructed firebase functions to track analytics within the application to seek areas of improvement and plan features
- Integrated Slack API to create a webhook directing help forms, removing the 1-hour delay between user and developer

Machine Learning Engineer Intern - Xerris Aug 2021

- Built technical documentation for clients and potential future developers outlining the 15 step instructions from preprocessing steps to model deployment through a combination of Jupyter Notebook and AWS Sagemaker
- Learned about Geometric Deep Learning and the math behind the concepts through manager-led training sessions
- Deconstructed machine learning python code into the various steps of the pipeline, we took into account how different variables throughout the code were storing training and test results to then be stored as a log of model progress

Software and Design Team Member - Robotics Team (FIRST Team 4733) Sep 2018 - Mar 2020

- Conceptualized the mechanical aspects of the block-carrying robot, problem-solving through 3 design requirements
- Monitored 300 lines of Java code for bugs on the software team, working directly with other members to resolve bugs
- Collaborated directly with team leads, actively learning by asking questions and observing the work done by all 5 teams

PROJECTS & COMPETITIONS

PC LED System - Personal Project Aug 2020

- Created a LED control system with 2 different colour cycling modes, either dynamic or static colour flows, for my PC with a C++ codebase controlling the LED strips in the case using an Arduino microcontroller, switches, and potentiometers
- Integrated system components within my PC case on a centralized breadboard with resistors and transistors
- Connected the system with the PC itself for power and to the 2 LED strips to project the appropriate colours

Basketball Mini Hoop Shot Counter - Arduino Project Oct 2020

- Created a basketball shot counting system with a greater than 95% accuracy rating using an Arduino Uno microcontroller, HC-SR04 (distance) and MPU-6050 (gyroscope) sensors, as well as Arduino libraries for the LCD screen displaying stats
- Implemented a 2-second delay to factor in the physics of the motion of basketball shots within the C++ codebase, this accounts for the delay of the ball hitting the rim and deflecting off various surfaces randomly before entering the rim

MIT COVID-19 Challenge - MIT Ideathon Nov 2020

- Oversaw the project by managing the creation of the idea and final presentation within a 36-hour window as team lead
- Presented all technical aspects of the project in a 5-minute presentation and answered any clarifying questions by judges
- Won 1st place for Systems involving Artificial Intelligence, out of a total of 20 other teams competing in the same track

EDUCATION

Candidate for B.A.S.c in Computer Engineering - University of Waterloo Sep 2021- Present